

Filter Summary Report: CG,TIA,Automated,NO,L,simple,Z4,Z5

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Contents

1 Examined $H(z)$ for CG TIA Automated NO L simple Z4 Z5: $\frac{Z_4Z_5g_m-Z_4}{2Z_4g_m+2Z_5g_m+2}$

$$H(z)=\frac{Z_4Z_5g_m-Z_4}{2Z_4g_m+2Z_5g_m+2}$$

2 AP

3 BP

4 BS

5 GE

6 HP

7 LP

8 X-INVALID-NUMER

8.1 X-INVALID-NUMER-1 $Z(s)=\left(\infty,\infty,\infty,\frac{1}{C_4s},\frac{R_5}{C_5R_5s+1},\infty\right)$

$$H(s)=\frac{-C_5R_5s+R_5g_m-1}{2C_4C_5R_5s^2+2g_m+s(2C_4R_5g_m+2C_4+2C_5R_5g_m)}$$

Parameters:

Q: $\frac{\sqrt{C_4}\sqrt{C_5}\sqrt{R_5}\sqrt{g_m}}{C_4R_5g_m+C_4+C_5R_5g_m}$
wo: $\frac{\sqrt{g_m}}{\sqrt{C_4}\sqrt{C_5}\sqrt{R_5}}$
bandwidth: $\frac{C_4R_5g_m+C_4+C_5R_5g_m}{C_4C_5R_5}$
K-LP: $\frac{R_5g_m-1}{2g_m}$
K-HP: 0
K-BP: $-\frac{C_5R_5}{2C_4R_5g_m+2C_4+2C_5R_5g_m}$
Qz: None
Wz: None

8.2 X-INVALID-NUMER-2 $Z(s)=\left(\infty,\infty,\infty,\frac{R_4}{C_4R_4s+1},\frac{1}{C_5s},\infty\right)$

$$H(s)=\frac{-C_5R_4s+R_4g_m}{2C_4C_5R_4s^2+2g_m+s(2C_4R_4g_m+2C_5R_4g_m+2C_5)}$$

Parameters:

Q: $\frac{\sqrt{C_4}\sqrt{C_5}\sqrt{R_4}\sqrt{g_m}}{C_4R_4g_m+C_5R_4g_m+C_5}$
wo: $\frac{\sqrt{g_m}}{\sqrt{C_4}\sqrt{C_5}\sqrt{R_4}}$
bandwidth: $\frac{C_4R_4g_m+C_5R_4g_m+C_5}{C_4C_5R_4}$
K-LP: $\frac{R_4}{2}$
K-HP: 0
K-BP: $-\frac{C_5R_4}{2C_4R_4g_m+2C_5R_4g_m+2C_5}$
Qz: None
Wz: None

8.3 X-INVALID-NUMER-3 $Z(s) = \left(\infty, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_4 R_5 s + R_4 R_5 g_m - R_4}{2C_4 C_5 R_4 R_5 s^2 + 2R_4 g_m + 2R_5 g_m + s(2C_4 R_4 R_5 g_m + 2C_4 R_4 + 2C_5 R_4 R_5 g_m + 2C_5 R_5) + 2}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{C_4} \sqrt{C_5} \sqrt{R_4} \sqrt{R_5} \sqrt{R_4 g_m + R_5 g_m + 1}}{C_4 R_4 R_5 g_m + C_4 R_4 + C_5 R_4 R_5 g_m + C_5 R_5} \\ \text{wo: } & \frac{\sqrt{R_4 g_m + R_5 g_m + 1}}{\sqrt{C_4} \sqrt{C_5} \sqrt{R_4} \sqrt{R_5}} \\ \text{bandwidth: } & \frac{C_4 R_4 R_5 g_m + C_4 R_4 + C_5 R_4 R_5 g_m + C_5 R_5}{C_4 C_5 R_4 R_5} \\ \text{K-LP: } & \frac{R_4 R_5 g_m - R_4}{2R_4 g_m + 2R_5 g_m + 2} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & -\frac{C_5 R_4 R_5}{2C_4 R_4 R_5 g_m + 2C_4 R_4 + 2C_5 R_4 R_5 g_m + 2C_5 R_5} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.4 X-INVALID-NUMER-4 $Z(s) = \left(\infty, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_4 g_m + s(C_5 R_4 R_5 g_m - C_5 R_4)}{2g_m + s^2(2C_4 C_5 R_4 R_5 g_m + 2C_4 C_5 R_4) + s(2C_4 R_4 g_m + 2C_5 R_4 g_m + 2C_5 R_5 g_m + 2C_5)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{C_4} \sqrt{C_5} \sqrt{R_4} R_5 g_m \sqrt{\frac{g_m}{R_5 g_m + 1}} + \sqrt{C_4} \sqrt{C_5} \sqrt{R_4} \sqrt{\frac{g_m}{R_5 g_m + 1}}}{C_4 R_4 g_m + C_5 R_4 g_m + C_5 R_5 g_m + C_5} \\ \text{wo: } & \sqrt{\frac{g_m}{C_4 C_5 R_4 R_5 g_m + C_4 C_5 R_4}} \\ \text{bandwidth: } & \frac{\sqrt{\frac{g_m}{C_4 C_5 R_4 R_5 g_m + C_4 C_5 R_4}} (C_4 R_4 g_m + C_5 R_4 g_m + C_5 R_5 g_m + C_5)}{\sqrt{C_4} \sqrt{C_5} \sqrt{R_4} R_5 g_m \sqrt{\frac{g_m}{R_5 g_m + 1}} + \sqrt{C_4} \sqrt{C_5} \sqrt{R_4} \sqrt{\frac{g_m}{R_5 g_m + 1}}} \\ \text{K-LP: } & \frac{R_4}{2} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & \frac{C_5 R_4 R_5 g_m - C_5 R_4}{2C_4 R_4 g_m + 2C_5 R_4 g_m + 2C_5 R_5 g_m + 2C_5} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

9 X-INVALID-ORDER

9.1 X-INVALID-ORDER-1 $Z(s) = (\infty, \infty, \infty, R_4, R_5, \infty)$

$$H(s) = \frac{R_4 R_5 g_m - R_4}{2R_4 g_m + 2R_5 g_m + 2}$$

9.2 X-INVALID-ORDER-2 $Z(s) = \left(\infty, \infty, \infty, R_4, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_4 s + R_4 g_m}{2g_m + s(2C_5 R_4 g_m + 2C_5)}$$

9.3 X-INVALID-ORDER-3 $Z(s) = \left(\infty, \infty, \infty, R_4, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_4 R_5 s + R_4 R_5 g_m - R_4}{2R_4 g_m + 2R_5 g_m + s(2C_5 R_4 R_5 g_m + 2C_5 R_5) + 2}$$

$$\mathbf{9.4 \quad X-INVALID-ORDER-4} \quad Z(s) = \left(\infty, \infty, \infty, R_4, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{R_4 g_m + s (C_5 R_4 R_5 g_m - C_5 R_4)}{2g_m + s (2C_5 R_4 g_m + 2C_5 R_5 g_m + 2C_5)}$$

$$\mathbf{9.5 \quad X-INVALID-ORDER-5} \quad Z(s) = \left(\infty, \infty, \infty, \frac{1}{C_4 s}, R_5, \infty \right)$$

$$H(s) = \frac{R_5 g_m - 1}{2g_m + s (2C_4 R_5 g_m + 2C_4)}$$

$$\mathbf{9.6 \quad X-INVALID-ORDER-6} \quad Z(s) = \left(\infty, \infty, \infty, \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_5 s + g_m}{2C_4 C_5 s^2 + s (2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.7 \quad X-INVALID-ORDER-7} \quad Z(s) = \left(\infty, \infty, \infty, \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{g_m + s (C_5 R_5 g_m - C_5)}{s^2 (2C_4 C_5 R_5 g_m + 2C_4 C_5) + s (2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.8 \quad X-INVALID-ORDER-8} \quad Z(s) = \left(\infty, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, R_5, \infty \right)$$

$$H(s) = \frac{R_4 R_5 g_m - R_4}{2R_4 g_m + 2R_5 g_m + s (2C_4 R_4 R_5 g_m + 2C_4 R_4) + 2}$$

$$\mathbf{9.9 \quad X-INVALID-ORDER-9} \quad Z(s) = \left(\infty, \infty, \infty, R_4 + \frac{1}{C_4 s}, R_5, \infty \right)$$

$$H(s) = \frac{R_5 g_m + s (C_4 R_4 R_5 g_m - C_4 R_4) - 1}{2g_m + s (2C_4 R_4 g_m + 2C_4 R_5 g_m + 2C_4)}$$

$$\mathbf{9.10 \quad X-INVALID-ORDER-10} \quad Z(s) = \left(\infty, \infty, \infty, R_4 + \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 R_4 s^2 + g_m + s (C_4 R_4 g_m - C_5)}{s^2 (2C_4 C_5 R_4 g_m + 2C_4 C_5) + s (2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.11 \quad X-INVALID-ORDER-11} \quad Z(s) = \left(\infty, \infty, \infty, R_4 + \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{g_m + s^2 (C_4 C_5 R_4 R_5 g_m - C_4 C_5 R_4) + s (C_4 R_4 g_m + C_5 R_5 g_m - C_5)}{s^2 (2C_4 C_5 R_4 g_m + 2C_4 C_5 R_5 g_m + 2C_4 C_5) + s (2C_4 g_m + 2C_5 g_m)}$$

10 X-INVALID-WZ

10.1 X-INVALID-WZ-1 $Z(s) = \left(\infty, \infty, \infty, R_4 + \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_4 C_5 R_4 R_5 s^2 + R_5 g_m + s (C_4 R_4 R_5 g_m - C_4 R_4 - C_5 R_5) - 1}{2g_m + s^2 (2C_4 C_5 R_4 R_5 g_m + 2C_4 C_5 R_5) + s (2C_4 R_4 g_m + 2C_4 R_5 g_m + 2C_4 + 2C_5 R_5 g_m)}$$

Parameters:

Q: $\frac{\sqrt{C_4} \sqrt{C_5} R_4 \sqrt{R_5 g_m} \sqrt{\frac{g_m}{R_4 g_m + 1}} + \sqrt{C_4} \sqrt{C_5} \sqrt{R_5} \sqrt{\frac{g_m}{R_4 g_m + 1}}}{C_4 R_4 g_m + C_4 R_5 g_m + C_4 + C_5 R_5 g_m}$

wo: $\sqrt{\frac{g_m}{C_4 C_5 R_4 R_5 g_m + C_4 C_5 R_5}}$

bandwidth: $\frac{\sqrt{\frac{g_m}{C_4 C_5 R_4 R_5 g_m + C_4 C_5 R_5}} (C_4 R_4 g_m + C_4 R_5 g_m + C_4 + C_5 R_5 g_m)}{\sqrt{C_4} \sqrt{C_5} R_4 \sqrt{R_5 g_m} \sqrt{\frac{g_m}{R_4 g_m + 1}} + \sqrt{C_4} \sqrt{C_5} \sqrt{R_5} \sqrt{\frac{g_m}{R_4 g_m + 1}}}$

K-LP: $\frac{R_5 g_m - 1}{2g_m}$

K-HP: $-\frac{R_4}{2R_4 g_m + 2}$

K-BP: $\frac{C_4 R_4 R_5 g_m - C_4 R_4 - C_5 R_5}{2C_4 R_4 g_m + 2C_4 R_5 g_m + 2C_4 + 2C_5 R_5 g_m}$

Qz: None

Wz: $\frac{\sqrt{-R_5 g_m + 1}}{\sqrt{C_4} \sqrt{C_5} \sqrt{R_4} \sqrt{R_5}}$

11 X-PolynomialError